

K BALAJI

192324180

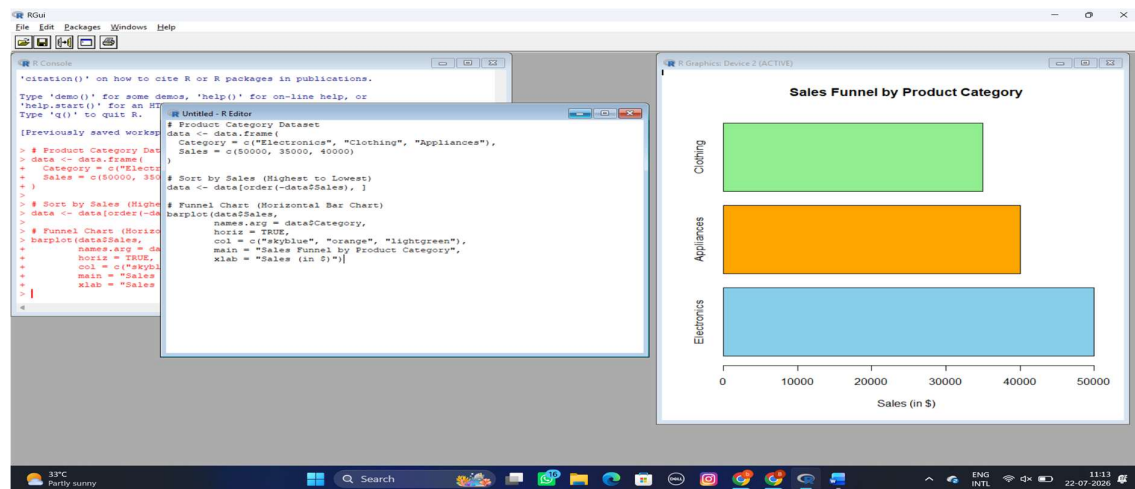
DATA HANDLING AND VISUVALIZATION LAB PROGRAMMS 11-15:

EXPERIMENT-11:

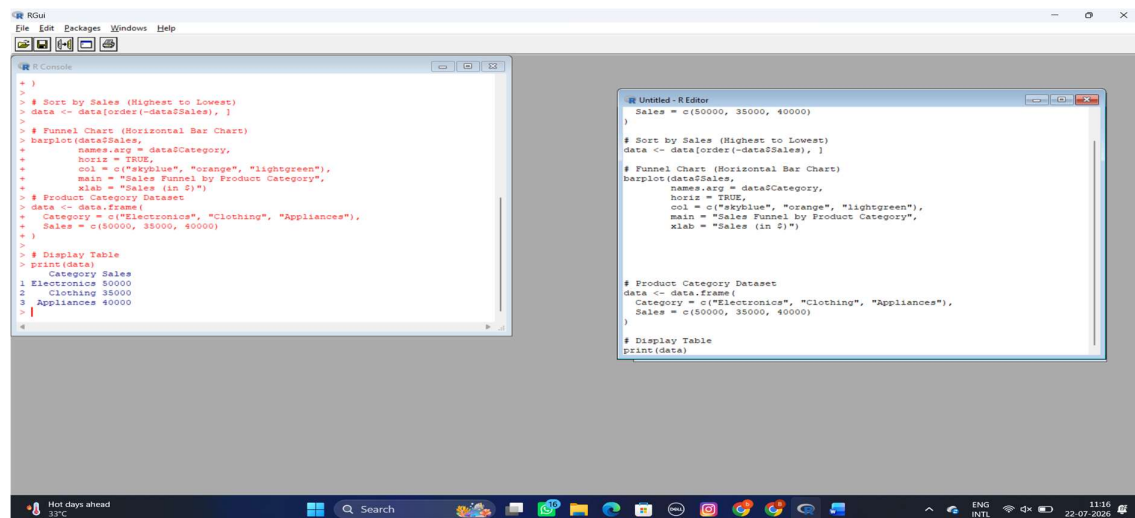
Product Category Analysis

Dataset: Product Categories

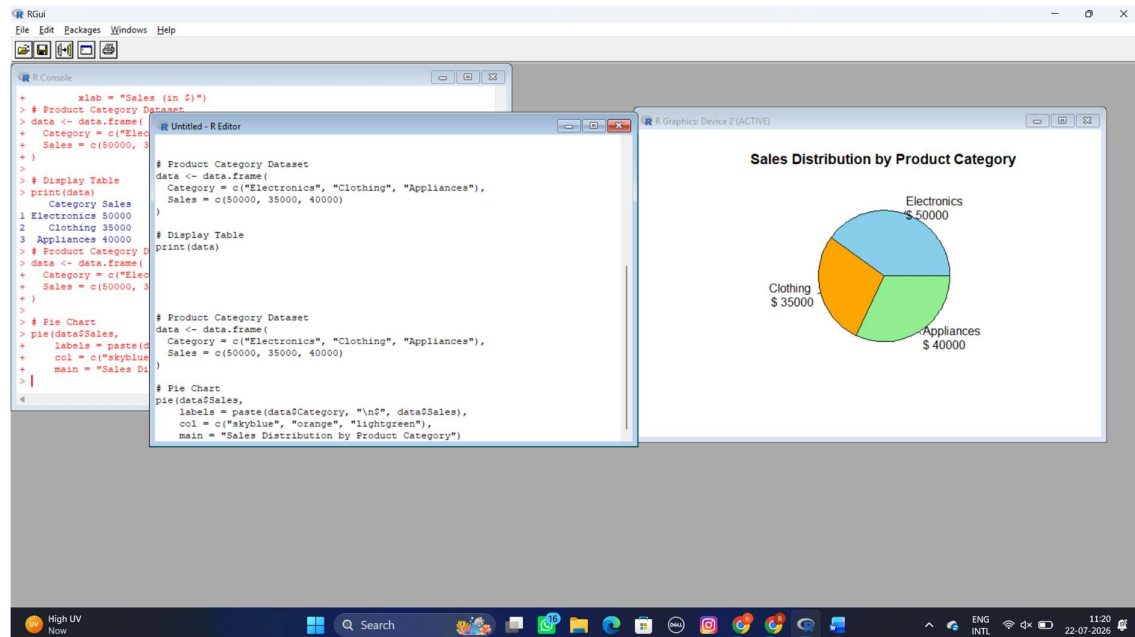
1.Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.



2) Build a table to display the sales data for each product category. Label the table elements.



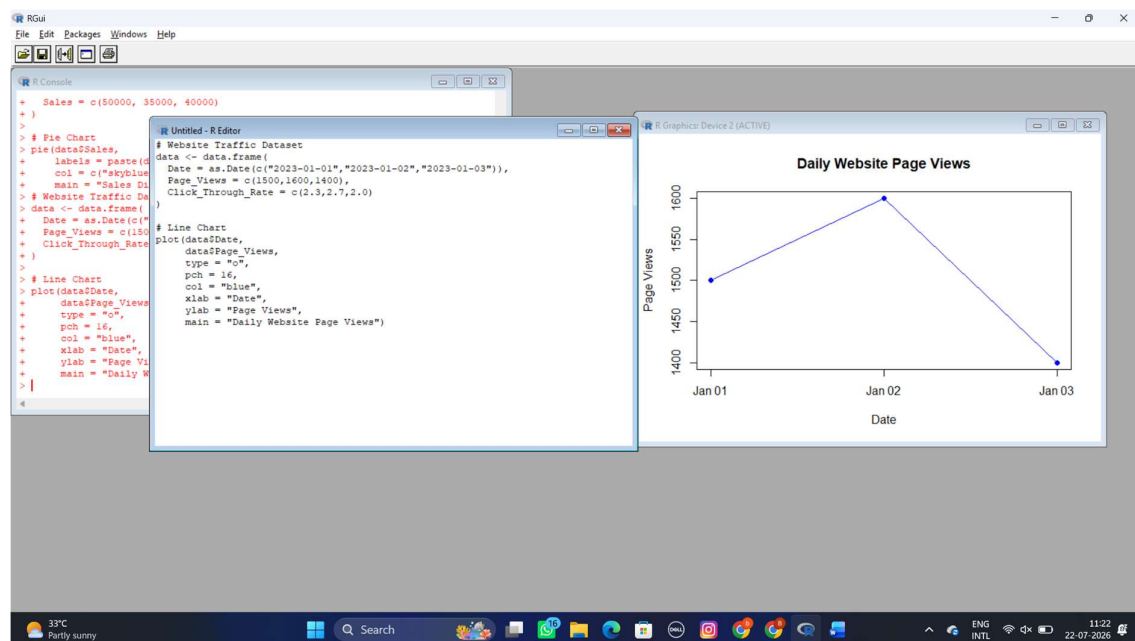
3) Create a pie chart to represent the distribution of sales across product categories. Include labels.



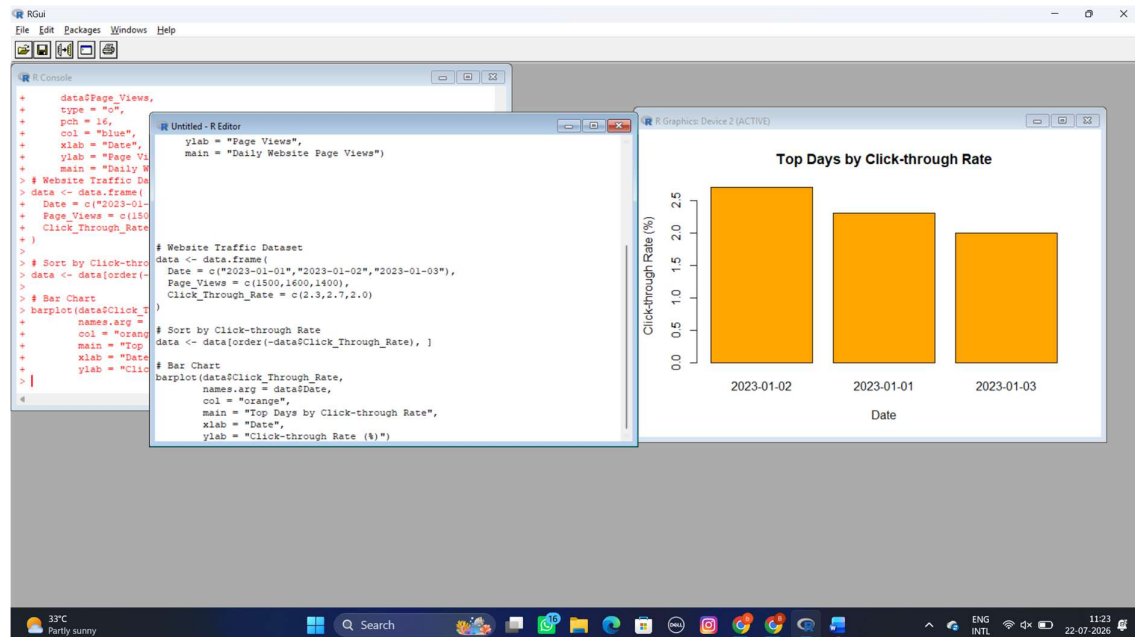
EXPERIMENT-12:

Dataset: Website Traffic

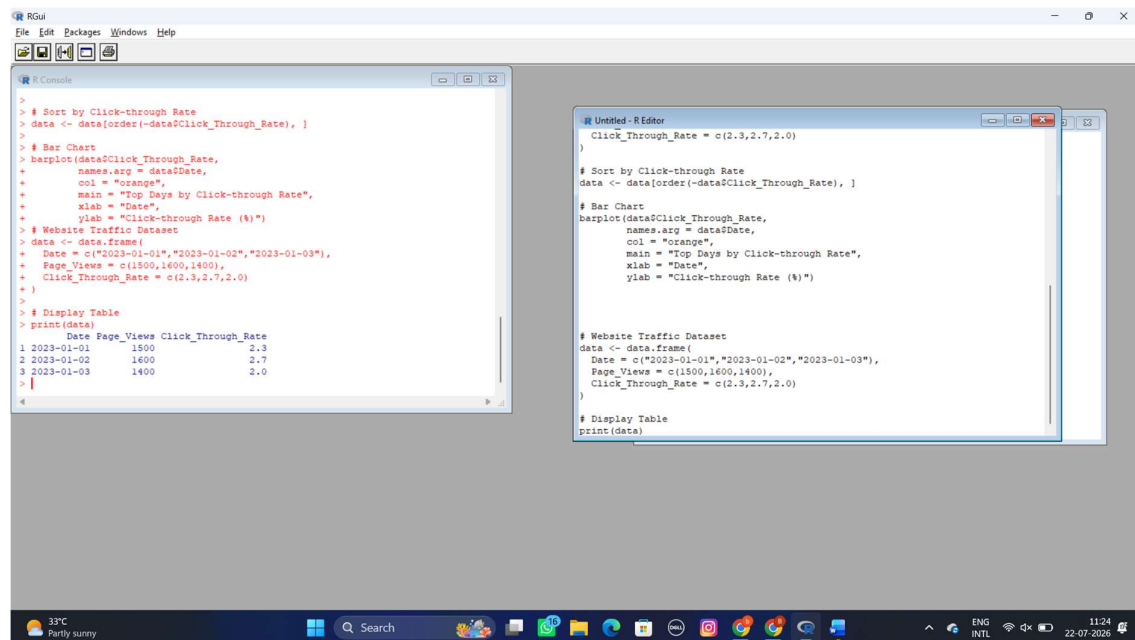
1) Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.



3. Build a table to show the website traffic data for each day. Label the table elements.

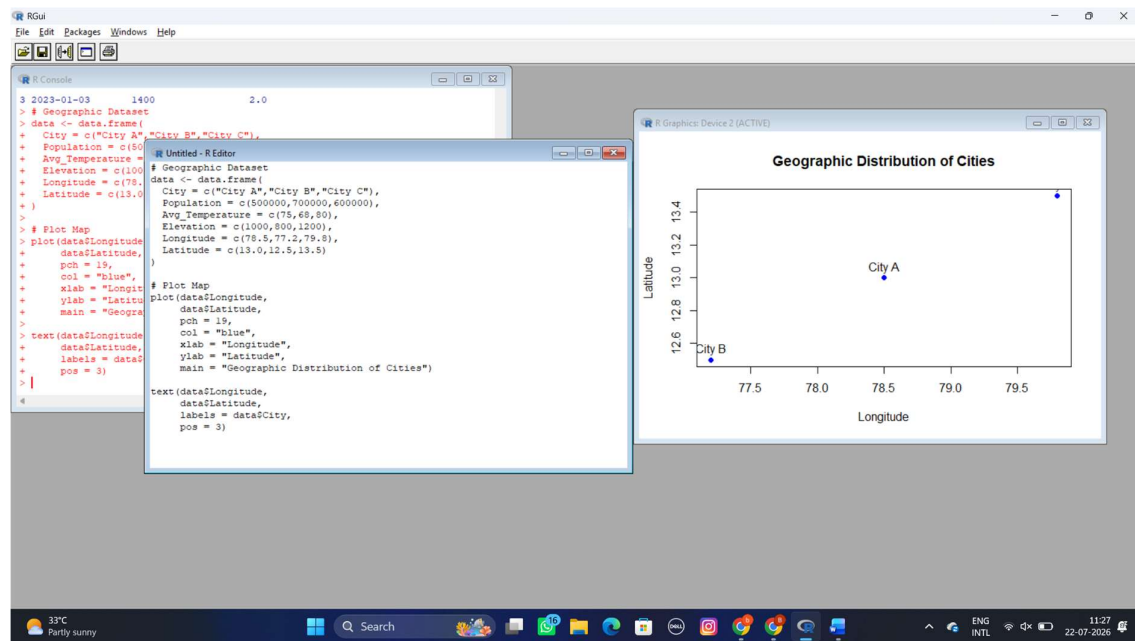


EXPERIMENT 13:

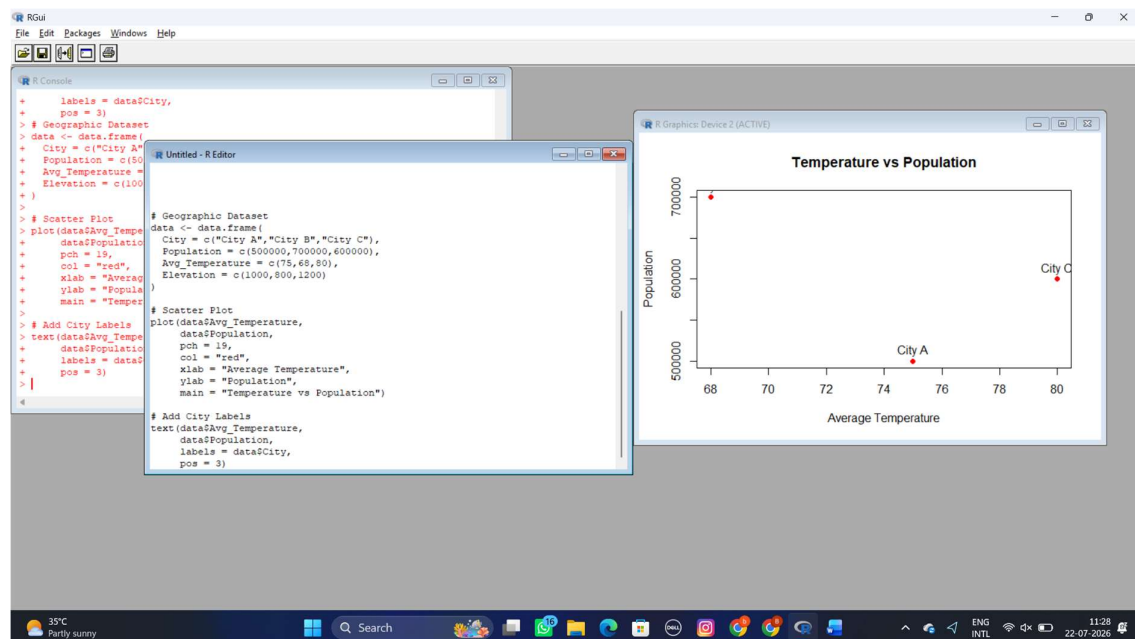
Dataset: Geographic Data

City Population Avg. Temperature Elevation

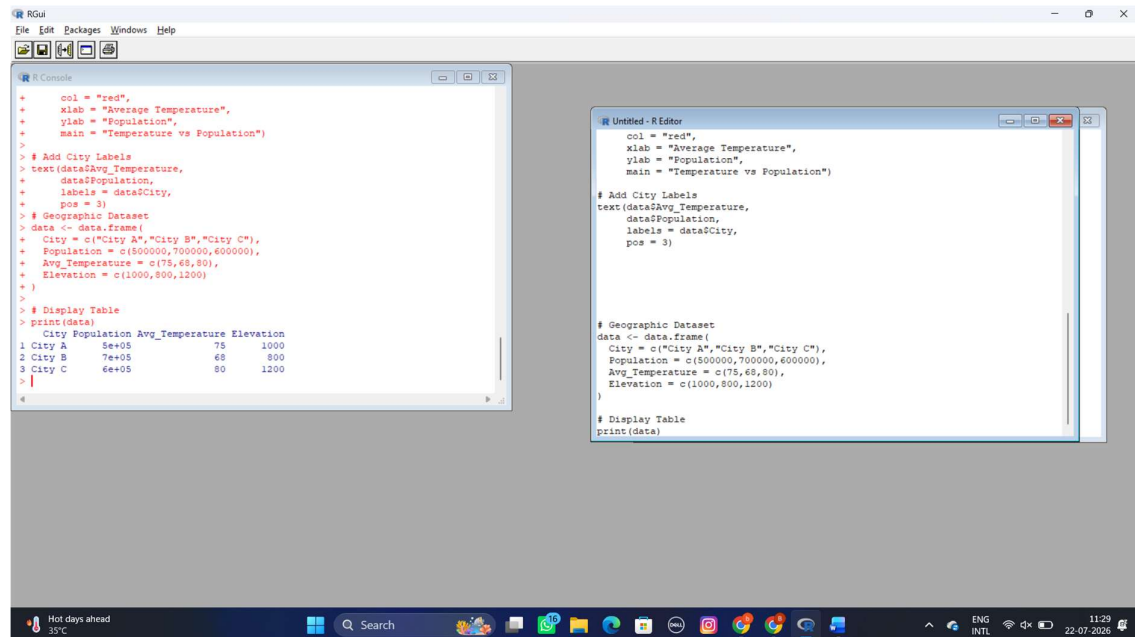
1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.



3. Build a table to display the geographic data for each city. Label the table elements.



The screenshot shows the RGui interface. The R Console window contains the following code:

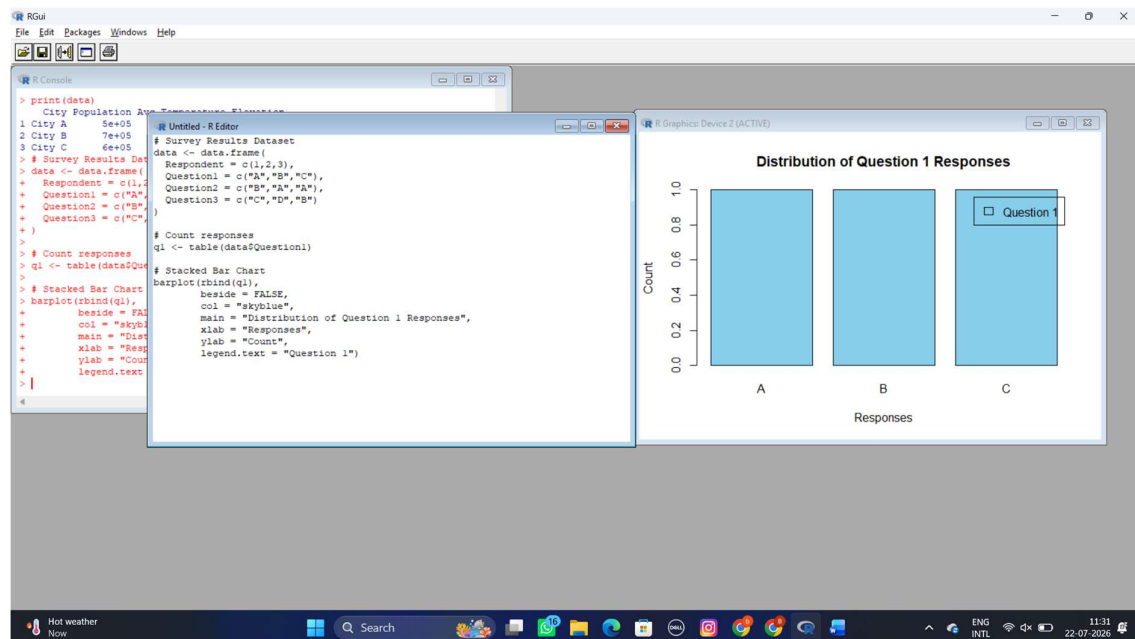
```
# col = "red",
# xlab = "Average Temperature",
# ylab = "Population",
# main = "Temperature vs Population")
>
# Add City Labels
> text(data$Avg_Temperature,
# data$Population,
# labels = data$City,
# pos = 3)
# Geographic Dataset
> data <- data.frame(
# City = c("City A", "City B", "City C"),
# Population = c(500000, 700000, 600000),
# Avg_Temperature = c(75, 68, 80),
# Elevation = c(1000, 800, 1200)
# )
>
# Display Table
> print(data)
  City Population Avg_Temperature Elevation
1 City A      5e+05           75      1000
2 City B      7e+05           68       800
3 City C      6e+05           80      1200
>
```

The R Editor window shows the same code as the R Console window.

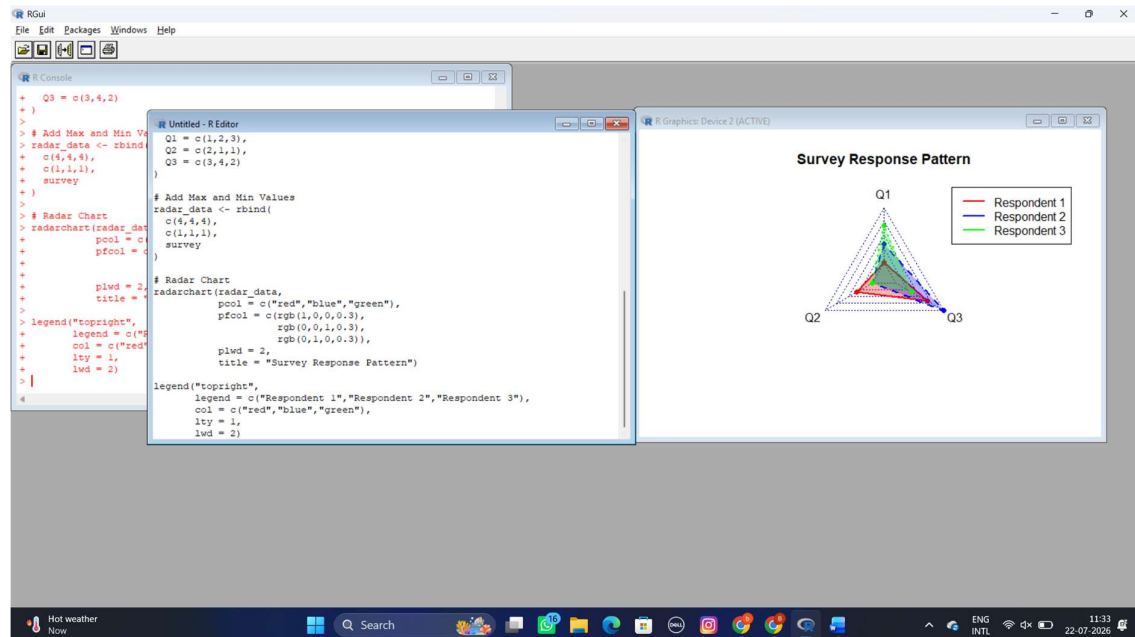
EXPERIMENT 14:

14. Dataset: Survey Results

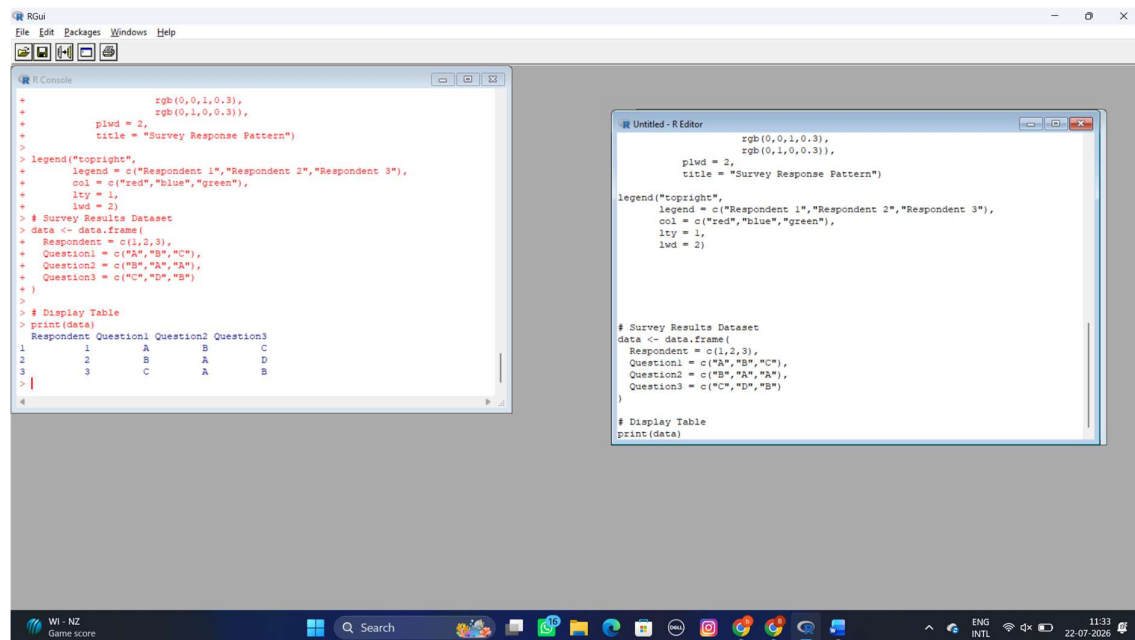
1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.



3. Build a table to show the survey response data for each respondent. Label the table elements.

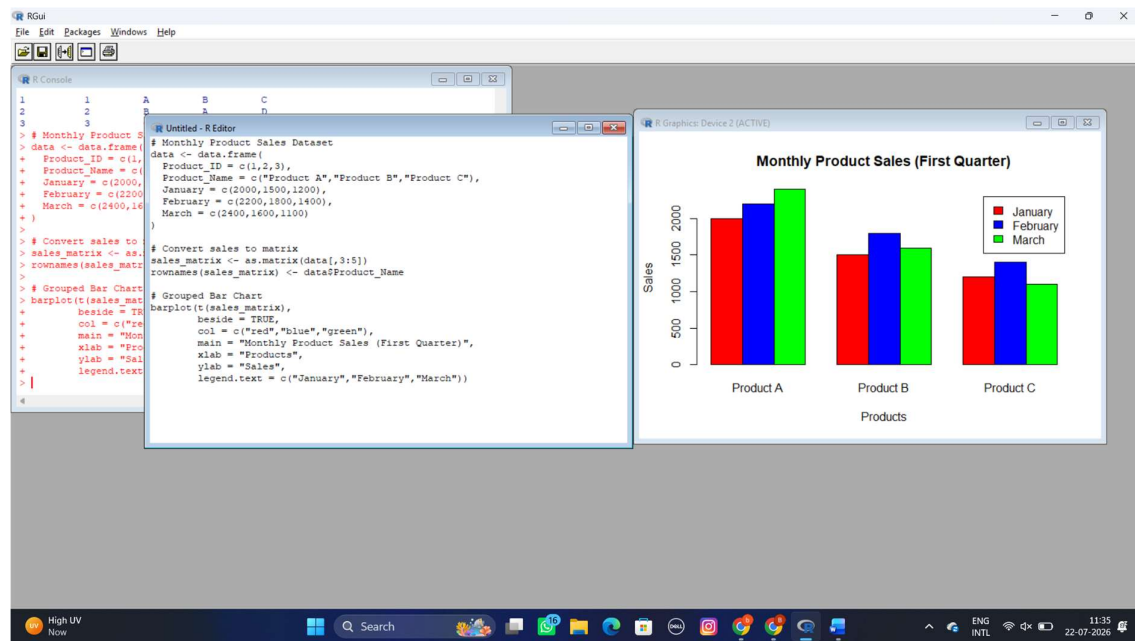


EXPERIMENT 15:

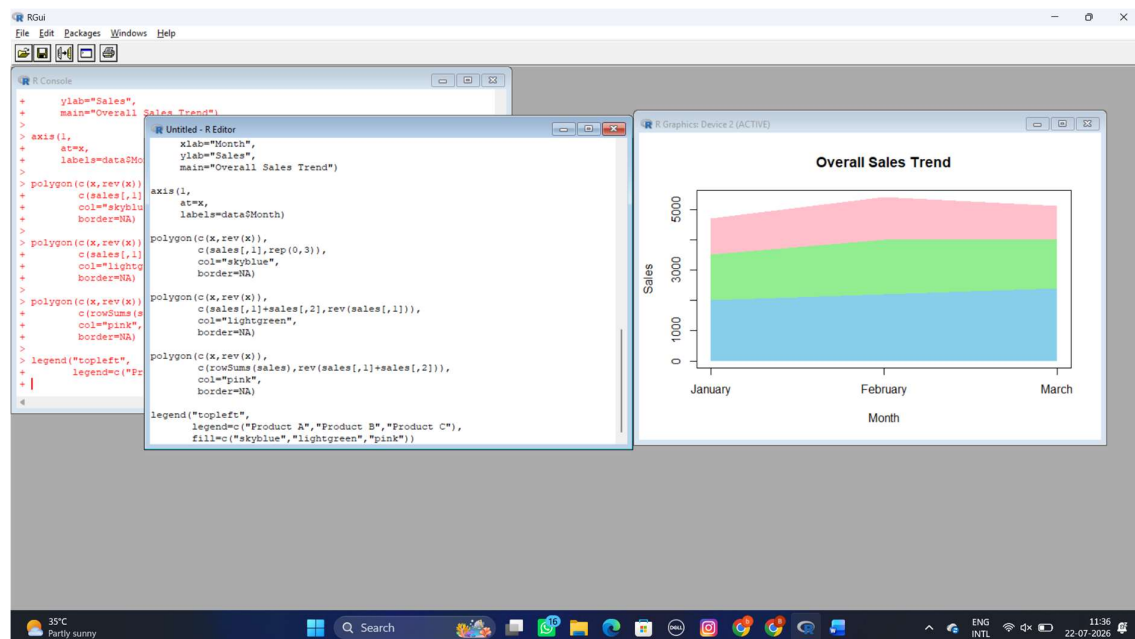
Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.

